

IILM UNIVERSITY

WEB DEVELOPMENT INTERNSHIP

# E-Commerce Platform & Project Management Dashboard

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|--------------------|-----------------------------------------------|
| STUDENT NAME       | Kuljeet Nischal                               |
| ENROLLMENT NO.     | 25SCS1003001305                               |
| PROGRAM & SEMESTER | B.Tech CSE (AI & ML) — 2nd Year, 3rd Semester |
| ORGANIZATION       | CodSoft                                       |

# Internship Overview

A one-month virtual Web Development internship at CodSoft



## POSITION

Web Development  
Virtual Internship



## DURATION

1 Month  
05 Aug – 05 Sep 2026



## OFFER ID

BY26RY226893  
ISO 9001:2015 Certified



## TRACK SELECTED

Level 3 — Advanced  
Full-Stack Track

## About CodSoft

CodSoft is a tech education and software development platform that runs virtual internships, project-based learning tracks, and AI-driven career tools — helping aspiring developers turn theoretical knowledge into practical, hands-on expertise.

# Task Selection

CodSoft requires completion of any ONE level — Level 3 (Advanced) was selected

## Level 1

FOUNDATIONAL

Portfolio · Landing Page · Calculator

## Level 2

INTERMEDIATE

Job Board · Online Quiz Maker

## Level 3

ADVANCED (SELECTED)

E-Commerce Website · Project Management Tool





LEVEL 3 · TASK 1

# E-Commerce Website

A full-stack online storefront — “Aillith” — where users can browse a live product catalog, search and filter items, view detailed product pages, manage a shopping bag, and complete checkout. A dedicated Add Product form lets an admin extend the catalog directly from the UI.

## User Flow



Browse &  
Search



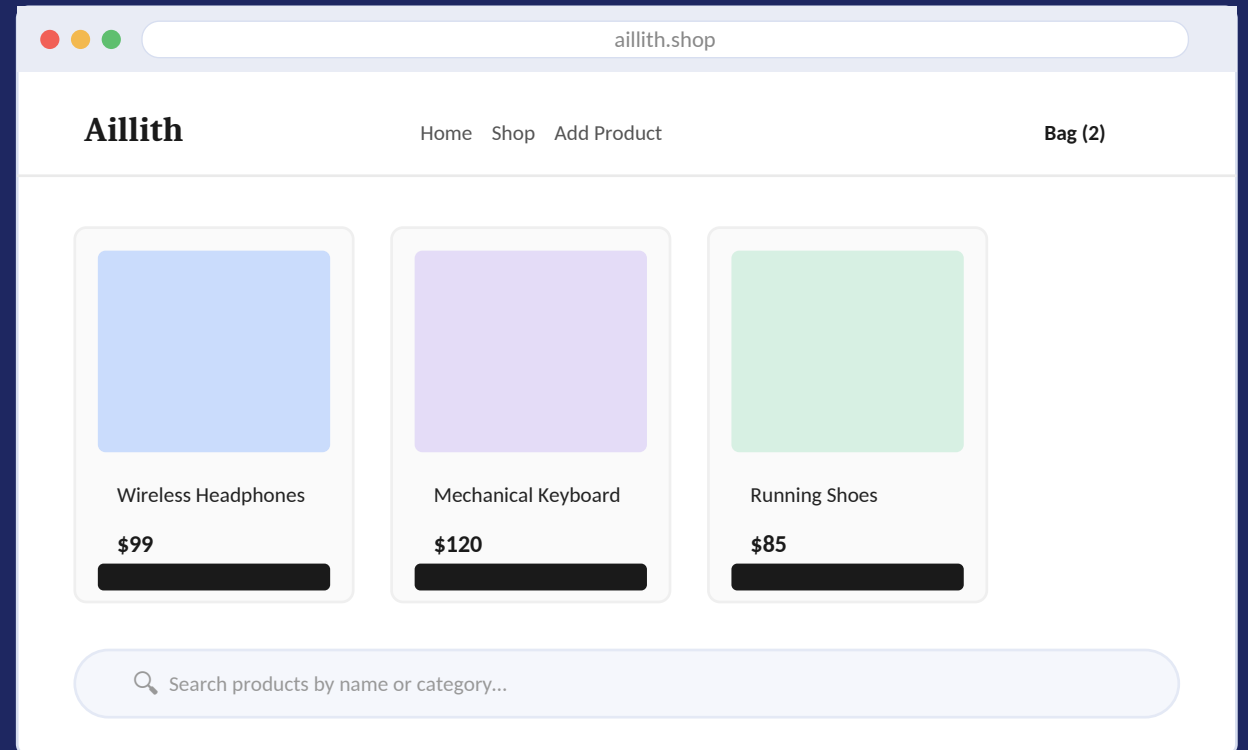
Product  
Details



Add to Bag



Checkout



# Task 1 — Key Features



## Live Product Catalog

Products fetched from a MongoDB collection via REST; catalog auto-seeds with sample items on first run.



## Search & Filter

Real-time client-side search across product name and category as the user types.



## Product Detail Page

Dynamic route per product (/product/:id) showing price, description, and an Add to Cart action.



## Shopping Bag & Checkout

Cart state managed in React; running total calculation and a confirmation screen on checkout.



## Add Product Form

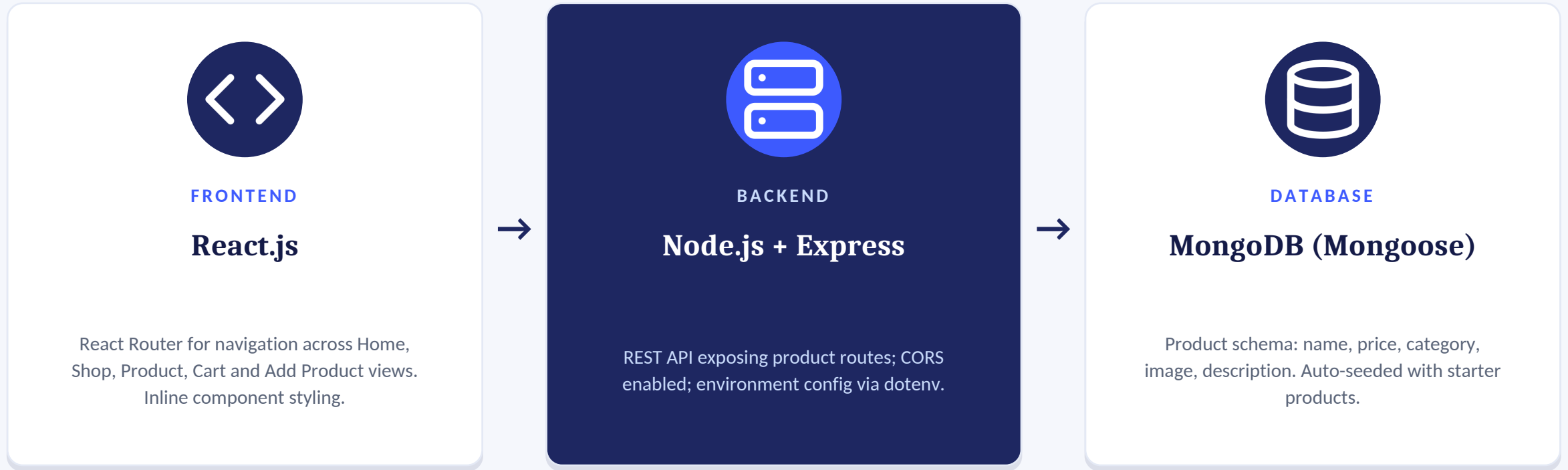
Admin-style form (name, price, category, image URL, description) that POSTs new items to the backend.



## REST API Backend

Express routes (GET/POST /api/products, GET /api/products/:id) backed by a Mongoose schema.

# Task 1 — Architecture & Tech Stack



## API Endpoints

GET /api/products

GET /api/products/:id

POST /api/products

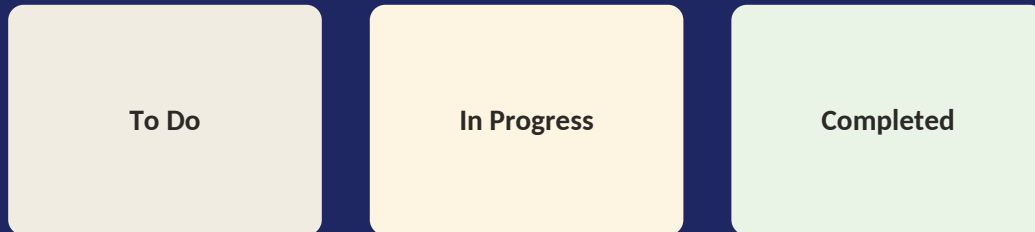


LEVEL 3 • TASK 2

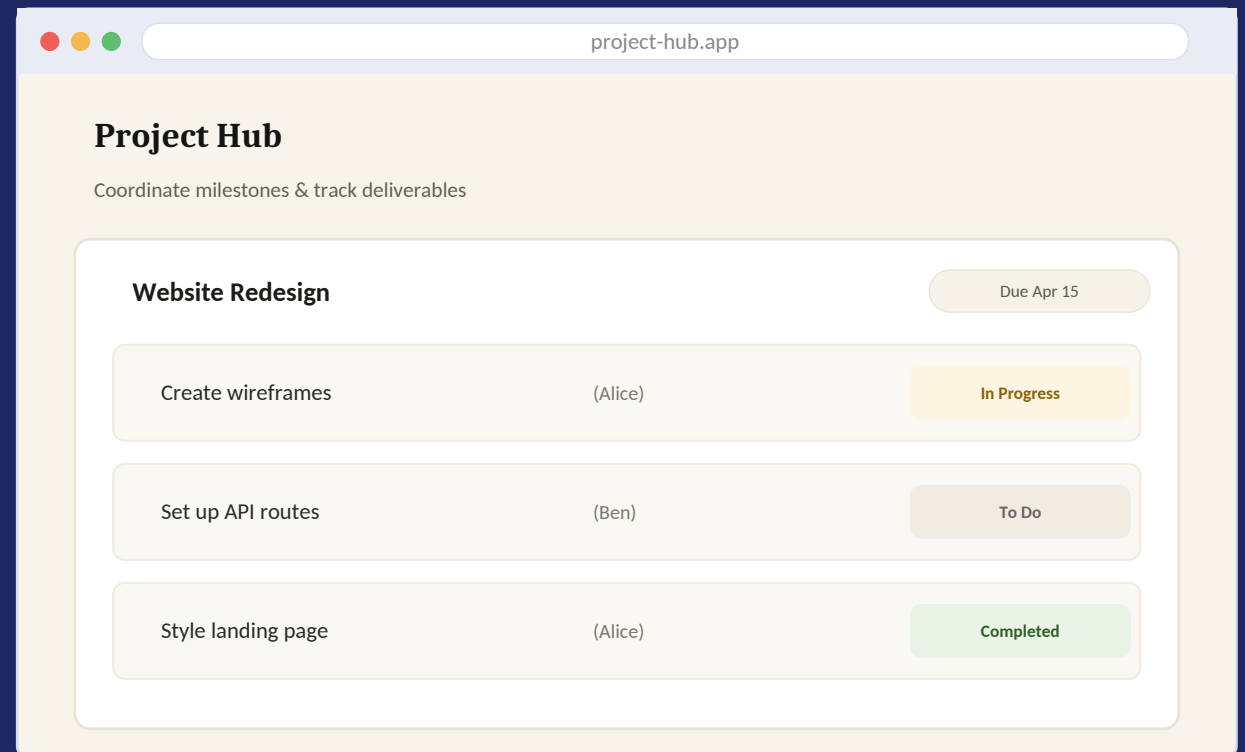
# Project Management Tool

“Project Hub” — a minimalist, cream-and-black styled dashboard for coordinating projects and tasks. Users can initialize new projects with a title, description and deadline, then break each project into tasks assigned to team members and tracked through a live status pipeline.

## Task Status Pipeline



*Updated live via a dropdown on each task card*



*Live mockup — reflects the actual "Project Hub" dashboard UI*

# Task 2 — Key Features



## Project Initialization

Create a project with title, description and deadline through a single form; each project renders as its own card.



## Task Breakdown

Every project holds a list of tasks, each with a title, assignee, deadline and status.



## Live Status Updates

Task status (To Do / In Progress / Completed) is changed instantly through a dropdown, with color-coded badges.



## Progress Overview

A running count of active projects and per-project deadline chips keep priorities visible at a glance.



## Task & Project Deletion

Individual tasks or entire projects can be removed directly from the dashboard.



## Lightweight REST API

Express routes handle project/task CRUD (create, update status, delete) over an in-memory data store.

# Overall Tech Stack

Technologies used across both Level 3 projects



**React.js**

Frontend UI, component state, routing



**Node.js / Express**

REST API server & routing logic



**MongoDB / Mongoose**

Persistent product data (Task 1)



**REST APIs**

JSON communication between client & server



**Custom CSS**

Responsive, hand-styled UI (no framework)



**Git & GitHub**

Version control & task submission

# Learning Outcomes

Skills gained through the internship

-  Building full-stack applications by connecting a React frontend to an Express/Node.js backend over REST APIs.
-  Designing and querying a MongoDB schema (Mongoose) for real, persisted application data.
-  Managing client-side state (cart contents, form inputs, task/project lists) using React Hooks.
-  Structuring multi-page applications with React Router and component-based architecture.
-  Implementing CRUD workflows end-to-end — create, read, update (status), and delete — across two independent apps.
-  Using Git/GitHub professionally: organizing a monorepo, structured commits, and public submission via repository links.



# Thank You

Both Level 3 tasks are complete and published on GitHub

`github.com/kuljeetnischal/CODSOFT_TASKSNO`

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